

Part 1: Privacy & Data Lifecycle Analysis of First Assessment

1.1 Data Minimisation Audit

In the pass task 1 I just submit, I made a security device using the sensor PIR motion sensor and accelerometer. When I'm testing my device I captured accelerometer x/y/z and PIR motion 5 times every second.

Before deciding this sample rate out in my program, I have thought about at what rate is the capture is a waste or too much. And I decided to use 5 times per second because in my case my device is use for preventing the user item being touch or has anyone entry the area. This touching or entering activity can be happened very fast, therefore we need a high detect frequency to check if anyone touch user item. However there might be some waste since normally if a person want to enter or touch the item or room, they still need to open the door or enter the area. During these extra step people it cannot touch item that fast. So I think there are still some waste of the detection time.

For sampling frequency, I think using 1 time per sec will be better for long run without losing patterns. For 5Hz = 18,000 row data and 1Hz = 3,600 row data.

I collect for around an hour and I think 10 minutes would have the same insight. As long as someone get detected by my device or move it. however changing from 5Hz to 1Hz would miss some detail like if someone briefly move my device, it might not able to detect. Because 1 sec gap might completely miss the movement.

1.2 Data retention & Lifecycle

I have made a python code to let my data store in local CSV file on my computer. For current stage, only I can access the csv file or people who have my computer password. The csv file is still in my computer. Looking at my csv my timestamp look like “0:54:24”, I didn’t include the real day in there so it hard to re-identify. However there are still a way to find out if the hacker is from the project related staff. Because they know what date the testing is on, so they can use timestamp to find out who is in this area at that time.

1.3 Ethical Stakeholders & Consent

When I am recording the data, me and my family data are captured. I did told them I am doing the project that will capture human movement and they all know. If the csv file misuse, the hacker will know what time we will be at home and what time we are out in the detected area, so they know our living pattern and might break in our house without anyone knowing.

privacy notice

"This area uses motion sensors and accelerometer monitors to secure personal belonging . Data is anonymised (no names, IDs). Raw readings are retained for 7 days, then aggregated to daily averages and retained for 90 days. Room occupants can request data deletion. Facility managers will receive weekly summaries only. Your comfort feedback (optional) is used to refine thresholds.

Part 2: Literature Review & Research Gap Identification

2.1 Literature search

A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone

<https://onlinelibrary.wiley.com/doi/full/10.1155/2018/6569826>

Smart Home Anti-Theft System:

<https://www.mdpi.com/2571-5577/1/4/42>

2.2 Synthesis & Gap Analysis

Source	Problem Addressed	Methods Used	Data Capture	Key Finding	Limitation
Liu, Bang, Liu, Nianbo, Chen, Guihai, Dai, Xili, Liu, Ming, <i>A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone, Mobile Information Systems</i> , 2018, 6569826, 16 pages, 2018. https://doi.org/10.1155/2018/6569826	High cost of commercial vehicle anti-theft systems and tracking devices	• Velocity-Assisted Dead Reckoning (VA-DR) • Map-matching algorithms • LSTM neural network • Ad-hoc authentication	• Built-in smartphone IMU (accelerometer & gyroscope) • GPS /Wi-Fi signals • Driving history logs	Low-cost tracking achieves localization errors lower than visual sight range	Requires continuous power supply & struggles with prolonged GPS loss without distinct road turns
Pandya, S.; Ghayvat, H.; Kotecha, K.; Awais, M.; Akbarzadeh, S.;	● Inability to detect intruders in total	• IoT framework & smart video analytics.	● Real-time video feeds from standard	● Successfully detects intruders wearing facial	● Dependent on a stable internet/network

Gope, P.; Mukhopadhyay, S.C.; Chen, W. Smart Home Anti-Theft System: A Novel Approach for Near Real-Time Monitoring and Smart Home Security for Wellness Protocol. <i>Appl. Syst. Innov.</i> 2018, <i>1</i>, 42. https://doi.org/10.3390/asi1040042	darkness or when faces are covered (masks/s carves). ● Standard security systems only record incidents after they happen.	• Near real-time motion and human detection algorithms. • Masked-face detection techniques. • Low-light image enhancement processing.	CCTV cameras. ● Physical IoT sensors	coverings. ● Significantly reduces false alarms and sends near instant mobile alerts. ● Achieves reliable intruder detection even in zero-light/pitch-black environments.	connection. ● Processing speed depends on local cloud/hardw are capabilities. ● Video quality under extreme low-light can affect detection accuracy.
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2.3 Identify Your Research Gap

Gap 1: False Alarms from Normal Car Movement and No Way to Check Who Entered the Car

The system detects theft simply by feeling the car move through the phone's motion sensors.

Gap: The study does not combine this motion tracking with other simple checks like detecting if a car door was opened, if a window was smashed or if an authorized driver's phone/key is nearby. Because of this, normal everyday events like a tow truck moving the car, strong wind, heavy traffic vibrations or a family member borrowing the car can trigger false theft alerts. Without a way to double check if an actual break in happened inside the car, the system causes unnecessary panic.

Gap 2: Not Tested in Real-World Weather or multiple Lighting

The authors tested theft detection under controlled low light and face cover conditions. However, they didn't test how the system handles real-world disruptions like passing car headlights, flickering lights, dark shadows, heavy rain, fog, bugs crawling on the camera, or curtains blowing in the wind.

Gap: Right now, the system is only tested in quiet indoor rooms. Its ability to spot shapes already drops from 97% in normal light down to 66% in total darkness. Will the camera break down or keep failing if it has to deal with outdoor weather, changing light, or random moving objects like tree branches?

Gap 2: Needs Internet to Work and Has No Backup Plan If Offline

The system depends heavily on sending live video through the internet to a central cloud server to send alerts to the owner's phone.

Gap: The study never tests what happens if the internet is slow, weak, or cut off—or if a burglar uses a signal jammer. If the internet or power goes out, the system cannot process the video or save backup footage locally, meaning it completely fails to send alerts when you need them most.

Part 3: Enhanced Use Case Design with Hypotheses

3.1 Refined Problem Statement

Original Problem Statement:

How can we build an affordable anti-theft device using a PIR sensor and accelerometer to catch someone touching or moving personal belongings on a desk by triggering local lights and sound?

Refined Problem Statement:

Simple security devices often make mistakes and trigger **false alarms**. For example, the buzzer might go off just because someone walked past the desk or gently bumped the table. When false alarms happen all the time, people get annoyed and turn off the device.

Main Research Question:

How can we combine data from two sensors, the PIR motion sensor (which sees body heat) and the LSM6DS3 accelerometer (which feels physical shaking) so the device can tell the difference between normal room activity and actual theft while reducing false alarms by 40%?

3.2 Sensor & Data Capture Protocol Design

1. Hardware Configuration

- **PIR Sensor (Spatial Proximity):** Detects body heat and human movement near the monitored desk or personal area.
- **LSM6DS3 Accelerometer (Physical Kinematics):** Built into the Arduino Nano 33 IoT board to detect 3-axis physical motion (X, Y, Z axes), tilting or direct snatching of personal belongings.

2. Enhanced Data Capture Specifications

Sampling Rate (Reading Speed): 1 per second

Sampling 1 time per second reduces data volume to **3,600 rows per hour**.

Physical theft or laptop movement lasts several seconds, so 1 Hz clearly captures the movement pattern without filling up board memory or crashing the system.

Collection Period: Data is collected 1 day (comparing normal room activity against simulated theft tests) to evaluate performance during both quiet nights and busy daytime hours.

Storage: local CSV

Data fields:

Each data row records the following specific fields:

1. Timestamp: Time elapsed in milliseconds (ms).
2. PIR_State: Motion status (0 = No body heat motion, 1 = Motion detected).
3. Accel_X, Accel_Y, Accel_Z: Raw acceleration along the 3 axes.
4. Accel_Delta: Total movement change magnitude calculated to measure shaking/movement intensity.

Privacy:

No Biometric or Personal Data: The device only logs simple numbers (infrared heat states and 3-axis movement). No cameras, microphones or user names are used.

Data Cleanup: Raw log files are used strictly for testing the detection algorithm and deleted after the project is finished.

3.3 Hypotheses

H1: Using Two Sensors Together is Better

Statement:

Using both sensors at the same time (PIR motion + LSM6DS3 accelerometer) will

catch real theft over 95% of the time and cause very few false alarms (less than 5%) compared to using only one sensor.

Simple Explanation:

When used alone, the PIR sensor triggers false alarms when people just walk past. The accelerometer triggers false alarms when someone bumps the table. Checking both sensors together stops these mistakes.

How to Test It:

Try 50 test runs (25 fake theft attempts and 25 normal daily activities like walking past or working at the desk). Count how many false alarms happen when using one sensor vs. both sensors together.

H2: Short Bumps vs. Real Theft

Statement:

Small accidents (like typing or lightly bumping the desk) only shake the sensor for a fraction of a second (< 1 second), but real theft (grabbing and moving an item) moves the sensor for at least 2 or 3 seconds in a row.

Simple Explanation:

A quick desk bump happens too fast to trigger an alarm at 1 Hz reading speed. Stealing an item takes longer, so the sensor will see movement across multiple seconds in a row.

How to Test It:

Compare the saved data logs of quick table bumps against real laptop movement to check how many seconds the shaking lasts.

3.4 Theoretical Validation

A. Literature-Based Validation

Alignment with Existing Theory:

Our hypotheses align directly with established Multi-Sensor Data Fusion (MSDF) frameworks and physical security literature.

Literature Support:

PIR Limitations: Research shows single PIR sensors suffer from high false-positive rates due to convection currents, HVAC airflow, or direct sunlight.

Accelerometers: Motion science shows that when someone moves or steals an object, the physical movement changes the total movement numbers

Application:

Checking heat motion (PIR) and physical shaking (accelerometer) together, checking how long the movement lasts, and using a total movement math formula all follow proven science to stop false alarms.

Statistical Feasibility

Sample Size Support:

Collecting data at 1 Hz yields 3,600 readings per hour, which equals 86,400 readings per day

Oversampling vs. Statistical Needs:

Standard statistical tests (such as paired t-tests or logistic regression) only require $n \geq 30$. Our dataset has massive oversampling, providing more than enough statistical power to test detection accuracy ($> 95\%$) and false alarm rates ($< 5\%$).

Statistical Assumptions to Test:

Temporal Autocorrelation: Consecutive 1 Hz readings are correlated over time. We will check time-series dependence and apply rolling moving averages.

Normality & Linearity: Baseline idle sensor noise will be tested for normal distribution to set accurate trigger thresholds.

C. Sensor Validation

Both sensors are ideal for this task.

LSM6DS3 Accelerometer Fitness:

The sensor datasheet shows it measures up to $\pm 16g$ of movement with low noise. Stealing an item creates movements over $0.3g$, so the sensor is plenty sensitive to catch a thief.

PIR Sensor Fitness:

The PIR sensor sees movement up to 7 meters away.

The PIR datasheet states it needs 10 to 30 seconds to warm up when turned on. the code will automatically throw away the first 30 seconds (30 readings) after bootup so it does not trigger a fake start-up alarm.

D. Data Quality Checks (Planned)

If the computer disconnects briefly, missing data under 5% will be filled using the last known good reading (ffill). If more than 5% of data is missing in an hour, that mean entire hour of data will be thrown out.

Outlier Strategy is Any acceleration reading higher than 16g which is impossible for normal human desk movements or negative time values will be flagged as sensor errors. These bad readings (less than 1% of total data) will be deleted and replaced with average values.

3.5 Monitoring & Real-Time Notification Design

To elevate our security device from a simple local alarm into a smart live security system, I will design a real-time monitoring dashboard and alerting engine.

My Plan:

Platform	Arduino Cloud Dashboard
Widgets	Live PIR state (Occupancy), Acceleration Magnitude (Movement), Security Status (Safe/Alert)
Warning Alert	PIR motion + Item movement ($\geq 2s$) $\rightarrow$ LED + Email/Browser push notification
User Privacy	Clear desk notification sign + Zero camera/audio logging

3.6 Revised Data Lifecycle & Ethics

Data Minimisation

In my first testing in P1, I recorded sensor data **5 times every second (5 Hz)**. This created 18,000 rows of data every hour. I realized that collecting data this fast creates too much wasted data. In this updated design, I reduced the reading speed to **1 time per second (1 Hz)**. This cuts the data down to 3,600 rows per hour, saving computer memory while still catching multi-second theft movements. I also strictly limit the data collected to simple numbers: Timestamp, PIR_State, raw 3-axis acceleration (Accel_X, Accel_Y, Accel_Z), and Accel_Delta. The system collects zero personal names, student IDs, camera video, or sound recordings.

Resubmit Answer

1. How does the system tell the difference between the Owner and an Intruder?

Since the system does not use cameras or face scanners, it tells the owner and intruder apart using simple rules and timing.

Manual Turn-Off: The owner can turn off the alarm before touching their item (for example, by pressing a hidden switch, using a phone app or flipping a switch on the online dashboard). If the item moves while the alarm is still turned ON, the system knows it is an intruder.

This is the most simple and cost effective way to separate a theft and a owner.

2. How can PIR and Accelerometer replace Video/Cameras?

The research paper uses CCTV cameras and video software to spot intruders. system replaces expensive cameras by measuring **heat and physical movement** instead of taking pictures.

PIR can detect near human movement and accelerometer can check if the object being move or not. using these two sensor, they can replace difficult cameras detection which require complex code to track human activity and limitation of object movement. And camera can be affect by different lighting as well.

Data Retention Policy

Raw Data (0–7 Days): Raw 1 Hz CSV log files are stored on my password-protected computer for 7 days to test the detection program and verify that false alarms are reduced.

Daily Summaries (8–90 Days): After 7 days, all raw sub-second files are converted into daily average summaries (such as total trigger counts and peak movement levels). The detailed raw 1 Hz files are permanently deleted.

Final Erasure (After 90 Days): All remaining summary files are permanently erased from my computer after 90 days.

Occupant Consent

Because testing happens in home spaces with family members around, informed consent is very important. Unprotected motion logs could reveal daily family routines, showing hackers when our home is empty. To prevent privacy issues, the following clear sign will be placed next to the device:

PRIVACY NOTICE

This area uses motion sensors and accelerometer monitors to secure personal belongings. Data is anonymised (no names, IDs). Raw readings are retained for 7 days, then aggregated to daily averages and retained for 90 days. Room occupants can request data deletion. Facility managers will receive weekly summaries only. Your comfort feedback (optional) is used to refine thresholds.

Stakeholders & Power Dynamics

Who Benefits: Item owners gain security for their belongings. Room occupants benefit because combining two sensors reduces annoying false alarm noise from the buzzer.

Power Imbalances & Fixes: A power imbalance happens if housemates or room occupants feel secretly watched or tracked. We fix this by using zero cameras or microphones and placing a clear privacy sign near the device. Occupants can also ask to delete their test data at any time.

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Google Gemini (accessed July 2026): Used to ask research and summarize content for easier understanding.

Used to All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

AI Ginimi output:

<https://onlinelibrary.wiley.com/doi/full/10.1155/2018/6569826> summary:

This link leads to the open-access research paper "**A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone**" published in *Mobile Information Systems* (Wiley/Hindawi, 2018) by Bang Liu, Nianbo Liu, Guihai Chen, Xili Dai, and Ming Liu.

Core Concept: *PhoneInside*

Instead of buying expensive, specialized vehicle tracking hardware, the authors propose **PhoneInside** — a framework that repurposes an unused, old smartphone into an automated anti-theft and tracking unit.

Key Technical Features

- **Repurposed Hardware Setup:** An obsolete Android smartphone is mounted inside the car's body and connected to a car charger for continuous power.
- **Theft Detection & Alerts:** Uses internal sensors (accelerometer, gyroscope) alongside an ad-hoc authentication protocol to detect unauthorized vehicle movement. If theft is detected, it immediately sends real-time alert notifications and location updates to the owner.
- **Adaptive Multi-Source Localization:** Combines GPS, cellular/Wi-Fi positioning, and inertial sensors to ensure continuous tracking, even when satellite signals are blocked (e.g., in parking garages or tunnels).
- **Novel VA-DR (Velocity-Assisted Dead Reckoning):** Standard dead

reckoning accumulates positional drift over time. The paper introduces a VA-DR algorithm that uses map data and turn angles at intersections/curves to estimate velocity and correct trajectory errors dynamically.

Main Contributions & Results

1. **E-Waste Reuse:** Demonstrates a practical, low-cost way to reduce electronic waste while providing property protection.
2. **High Tracking Accuracy:** Experimental testing showed that the system tracks stolen vehicles accurately, keeping average localization errors well within visual line-of-sight range.
3. **Cost-Effective:** Achieves performance comparable to commercial OBD-II trackers or dedicated GPS units at a fraction of the cost.

1. Map-Matching Algorithms & Dead Reckoning

- **Velocity-Assisted Dead Reckoning (VA-DR):** Calculates trajectory without an OBD-II connection by combining inertial sensor data with street layout geometries.
- **Topological Map-Matching:** Aligns raw GPS/location points onto known digital road segments to eliminate drift.

2. Signal Processing & Classical State Estimation

- **Sensor Filtering:** Preprocesses raw accelerometer and gyroscope data to remove noise and gravity biases before calculating vehicle turns or movement.
- **Finite State Machine (FSM):** Manages system transitions between monitoring modes (e.g., *Parked*, *Vibration Detected*, *Theft Warning*).

3. Machine Learning (Deep Learning)

- **Long Short-Term Memory (LSTM) Network:** Learns the owner's driving history to build a custom "mobility model." It analyzes driving patterns to detect abnormal driving behavior characteristic of theft.

<https://www.mdpi.com/2571-5577/1/4/42> give me a easy summary what it about

This paper is titled "**Smart Home Anti-Theft System: A Novel Approach for Near Real-Time Monitoring and Smart Home Security for Wellness Protocol**".

What It's About in Simple Terms

1. **The Core Idea:** It proposes an **Internet of Things (IoT)-based smart home security system** designed to catch burglars in real time and immediately notify the homeowner on their device so action can be taken right away.
2. **The Problem with Standard Security Systems:**
 - Traditional CCTV systems usually just record video footage, meaning you only discover a theft **after** it has already happened.
 - Old systems struggle to tell the difference between humans and pets/objects, frequently trigger false alarms, or fail if an intruder covers their face with a mask, hat, or scarf.
3. **How This System Fixes It:**
 - **Detects Hidden Faces:** Uses smart detection algorithms to spot human intruders even if their face is partially or fully covered.
 - **Works in the Dark:** Can detect intruders in pitch-black conditions using standard CCTV cameras without needing expensive night-vision gear.
 - **Instant Alerts:** Combines physical sensors with cloud software to process video data quickly and send near-instant alerts to the homeowner's phone.

i am confuse, explain what i need to do in P2 Part3 in easy way.

What is Part 3 asking you to do?

Think of Part 3 as taking your simple P1 anti-theft security device (the PIR sensor + LSM6DS3 accelerometer) and designing a real, scientific test for it.

Instead of just saying *"I built a device that beeps when touched,"* Part 3 asks you to explain:

1. What **specific problem** you are trying to solve.
2. How you will **collect sensor data** properly.
3. What **educated guesses (hypotheses)** you are testing.
4. Why your plan makes sense **theoretically and ethically**.

Step-by-Step Breakdown (Made Easy)

1. Refined Problem Statement

- **What to do:** Take the problem from P1 and make it more specific.
- **Simple Idea:** "PIR sensors often trigger false alarms when people just walk past or when the heater turns on. How can we combine the PIR sensor and the LSM6DS3 accelerometer to catch actual theft while reducing false alarms by 40%?"

2. Sensor & Data Capture Protocol Design

- **What to do:** Fill out or write down the exact rules for how your device gathers data.
- **Simple Idea:**
 - **Which sensors?** PIR (detects body heat/proximity), LSM6DS3 (detects physical moving/shaking), and optional DHT22 (detects room temperature changes).
 - **How fast?** Sample PIR and IMU at **10 Hz** (10 times per second).
 - **For how long?** Collect data over 14 days (or simulated test runs).

- **What data?** Time, $X/Y/Z$ acceleration values, motion state (HIGH/LOW), and room ID.
- **Privacy:** Don't record personal names, camera video, or exact GPS locations.

3. Hypotheses (Your Educated Guesses)

- **What to do:** Write 2–3 clear statements that you can test with numbers (H_1, H_2, H_3).
- **Simple Idea:**
 - H_1 (**Main Guess**): Combining PIR motion *AND* accelerometer movement at the same time will be much more accurate ($> 95\%$) than using just one sensor alone.
 - H_2 (**Vibration Guess**): Someone bumping the desk makes a quick spike (< 200 ms), but someone stealing the laptop makes a continuous movement (> 600 ms). A moving average filter will ignore the quick bump.
 - H_3 (**Temperature Guess**): Sudden changes in air conditioning cause false PIR triggers when no one is in the room.

4. Theoretical Validation (Pre-Execution Checks)

- **What to do:** Prove that your setup makes sense *before* running the full test.
- **Simple Idea:**
 - Show the math formula you use to measure movement regardless of how the device is sitting on the desk:

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- Explain why you clean the data (e.g., throw away corrupt sensor readings or apply a moving average filter to reduce noise).

5. Monitoring & Real-Time Notification Design

- **What to do:** Draw or explain what happens when motion is detected.
- **Simple Idea:**
 - **Level 1 (Slight Motion):** Someone walks near → Blink a low LED warning.
 - **Level 2 (Theft / Heavy Movement):** Someone picks up the laptop → Sound the buzzer + send a push notification to your phone via Arduino IoT Cloud.

6. Revised Data Lifecycle & Ethics

- **What to do:** Explain how you keep people's data safe and private.
- **Simple Idea:**
 - Delete raw high-speed sensor data after 7 days.
 - Create a simple **Privacy Notice** sign to stick near the desk so room occupants know only motion and acceleration are logged (no audio or video).

Quick Checklist to Complete Part 3

1. **Define your problem clearly** (Focus on stopping false alarms using sensor fusion).
2. **Specify your data settings** (10 Hz sampling rate, 14 days, privacy protection).
3. **Write 2 to 3 testable hypotheses** (H_1, H_2, H_3).
4. **Explain your math/filters** (Moving average filter & magnitude formula).
5. **Describe alert levels** (Local light/sound vs. Remote phone alerts).
6. **Add an ethics notice** (Inform users what data is collected).